

Resonance Profiles and Sharp Endpoint Laws for Rotating Fabius–Rvachev Distributions

Uniform irrational asymptotics, phase-dependent quantiles,
and superlinear Liouville corrections

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Abstract

We investigate two explicit endpoint conjectures in the matrix-dilated Fabius–Rvachev report in the ProveIt repository. For independent uniform random variables, consider

$$S_\beta = \sum_{n \geq 1} 2e^{-\lambda n} |\cos \pi(n\alpha + \beta)| V_n, \quad V_n \sim \text{Unif}[0, 1], \quad \lambda > 0.$$

Every irrational rotation has the same leading logarithmic Laplace and small-deviation laws, uniformly in the phase; no Diophantine hypothesis is needed for this leading term. For a badly approximable rotation, we obtain an explicit phase-uniform refinement

$$-\log \mathbb{E} e^{-e^t S_\beta} = \frac{t^2}{2\lambda} - \frac{t}{2} - R_\beta(t) + O_{\alpha, \lambda}(\log^2(2+t)),$$

where R_β is a nonnegative, nondecreasing, 1-Lipschitz profile formed from the deepest truncated return to a cosine zero. This proves a stronger version of the displayed Diophantine estimate in the repository, including phases with an exact zero. A tilted gamma approximation supplies a relative saddle-point formula, a three-term logarithmic cap-probability expansion, and an explicit multiplicative inverse-quantile asymptotic.

The phase profile cannot in general be replaced by a bounded or logarithmic carrier merely by excluding exact zeros. Almost every phase has a common quantile asymptotic, whereas a dense G_δ set of nonexact phases, of Hausdorff dimension zero, has a quantile ratio with lower limit $q = e^{-\lambda}$ and upper limit 1. More generally, the limits are classified by an inhomogeneous exponential approximation exponent. Finally, for a dense G_δ set of Liouville rotations we construct, for every integer $k \geq 2$, Laplace-correction subsequences of order $t^{2-1/k}$. These resolve the repository's Liouville-spike question in a stronger, baseline-corrected form. All arguments are given in ordinary mathematical detail; numerical checks are supplementary, not proofs.

Keywords. Fabius function; small deviations; irrational rotation; Sudler products; exponential tilting; inverse quantiles; Liouville numbers.

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1 Research target, contribution boundary, and source status

1.1 The concrete questions being addressed

The repository report *Matrix-Dilated Fabius–Rvachev Laws* [1] constructs smooth laws on self-affine zonotopes and reduces their directional boundary probabilities to infinite products of uniform Laplace transforms. In its section on directional caps, the source labels `conj:diophantine-endpoint` and `conj:liouville` ask, respectively, for control of irrational-rotation endpoint corrections and for exceptional Liouville spikes. The first displayed conjectural estimate, in the source’s notation, is

$$\log \mathcal{L}_w(s) = -\frac{(\log s)^2}{2\lambda} + O(\log s \log \log s). \quad (1)$$

It assumes bounded continued-fraction type and a phase described as nonresonant. Its accompanying discussion suggests a bounded or logarithmically growing quasiperiodic correction. The second question asks whether sufficiently close Liouville returns force unbounded corrections.

We resolve the displayed estimate with a stronger $O(\log s)$ error, uniformly in *all* phases, and identify the additional term responsible for that error. We also prove that excluding exact zeros alone does not justify the suggested small carrier: exceptionally well approximated phases already give linear corrections for a fixed badly approximable rotation. The Liouville construction gives still larger, but subquadratic, corrections. We do *not* assert a full all-orders endpoint expansion or a bounded quasiperiodic residual under stronger phase hypotheses.

1.2 Relation to existing mathematics

The scalar infinitely differentiable compactly supported function and its probabilistic structure are classical; Arias de Reyna’s exposition [2] is a primary reference relevant to the repository’s scalar normalization. The products generated by $\log |2 \sin \pi(n\alpha + \gamma)|$ belong to the established theory of Sudler products and singular Birkhoff sums. Verschueren and Mestel [3] prove convergence along a Fibonacci subsequence and power-law bounds for the golden rotation. Borda [4] studies value distributions and concentration of Sudler products, together with Birkhoff sums for irrational rotations. Hauke [5] investigates how badly approximable continued fractions affect extreme Sudler-product behavior. These are related antecedents, not references establishing the present cap-distribution and inverse-quantile conclusions.

Our use of bounded-variation discrepancy, exponential tilting, gamma asymptotics, Borel–Cantelli, and Baire category is standard methodology. In particular, a useful trimmed $O(\log^2 N)$ rotation estimate below is proved directly; it is not presented as a new general theorem about Sudler products. The research contribution developed here is the explicit phase-uniform endpoint profile, its transfer through a relative saddle formula to inverse quantiles, the resulting phase classification, and the rational-shadowing construction for the source’s Liouville question.

1.3 What the provenance check does and does not establish

The source conjectures were inspected at repository commit

fbba58593dc0622aa914972896150d4848f935b5.

The source-file blob is 143f6da778dfc0d5ab04133005ba6715a3cf155b. Appendix A records the path and exact source labels. The audit was targeted: it inspected the matrix report, its relevant surrounding definitions, the repository manifest, and searches for the conjecture label and proposed profile terminology. It was not a proof-by-proof review of the entire repository. The search index and

pinned-file revision were not identical, so an unsuccessful phrase search is not treated as a certificate of repository-wide novelty.

All theorems in this article are supported by the proofs below rather than by assuming that the repository’s other research claims are correct. The cap product itself is rederived. The proofs have not been checked by a proof assistant or independent referee. The targeted literature check establishes context, not universal publication priority. Accordingly, “proved here” means a mathematical argument is supplied here; it does not mean that a priority claim has been independently certified.

Source question or issue	Result and remaining boundary
Leading irrational endpoint scale	Theorem 4.1: no bounded-type assumption is needed; convergence is phase-uniform.
Displayed Diophantine estimate	Theorem 6.2: an explicit profile and $O(\log^2 \log s)$ residual, with total error $O(\log s)$.
Boundary probabilities and inversion	Theorems 7.3–8.2: relative saddle formula and phase-corrected inverse asymptotic.
Meaning of “nonresonant”	Theorems 9.2–9.3: no exact zero is too weak for a bounded or logarithmic correction.
Liouville spikes	Theorem 10.2: corrections of order $(\log s)^{2-1/k}$ along subsequences, for every integer $k \geq 2$.
All-orders quasiperiodic expansion	Not proved; precise remaining questions are given in Section 12.

2 Rotating caps and a fixed normalization

Fix $\lambda > 0$, set $q = e^{-\lambda}$, and let $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}/\mathbb{Z}$. Unless a rational comparison is explicitly under discussion, α is irrational. Let $(V_n)_{n \geq 1}$ be independent random variables with distribution $\text{Unif}[0, 1]$. Define

$$a_n(\beta) = 2|\cos \pi(n\alpha + \beta)|, \quad b_n(\beta) = q^n a_n(\beta), \quad (2)$$

$$S_\beta = \sum_{n \geq 1} b_n(\beta) V_n, \quad C_\beta(x) = \mathbb{P}(S_\beta \leq x), \quad (3)$$

$$\mathcal{L}_\beta(s) = \mathbb{E} e^{-s S_\beta}, \quad H_\beta(t) = -\log \mathcal{L}_\beta(e^t). \quad (4)$$

The series converges for every sample point, since $\sum b_n \leq 2q/(1-q)$. All logarithms are natural.

Proposition 2.1 (Reduction of a directional cap). *Let U_n be independent $\text{Unif}[-1, 1]$ variables, let R_θ denote planar rotation through θ , and let*

$$X = \sum_{n \geq 1} q^n R_{n\theta} v U_n, \quad v \in \mathbb{R}^2 \setminus \{0\}.$$

For $w \neq 0$, put $h(w) = \sum_{n \geq 1} q^n |\langle w, R_{n\theta} v \rangle|$. If $\alpha = \theta/\pi$ and δ is the angle of v minus the angle of w , then

$$h(w) - \langle w, X \rangle \stackrel{d}{=} \|v\| \|w\| S_\beta, \quad \beta = \delta/\pi \pmod{1}.$$

Moreover,

$$\mathcal{L}_\beta(s) = \prod_{n \geq 1: b_n > 0} \frac{1 - e^{-s b_n}}{s b_n}, \quad s \geq 0. \quad (5)$$

A zero coefficient contributes a factor 1.

Proof. The support bound in direction w is the sum of the absolute projected segment lengths, and every finite choice of segment endpoints is approximable within the support, so its supremum is $h(w)$. For

a real coefficient c , $|c| - cU$ has the same law as $2|c|V$. Apply this independently to every projected term and use $\langle w, R_{n\theta}v \rangle = \|v\|\|w\| \cos(n\theta + \delta)$. The finite-product formula follows by integration of a uniform variable. Bounded convergence of the exponential as the finite sum tends to S_β gives the infinite-product identity. $\square$

The factor $\|v\|\|w\|$ is not silently absorbed into λ or β . For a physical deficit $D = AS_\beta$ with $A > 0$, substitute $s \mapsto As$ in the Laplace formulas and $x \mapsto x/A$ in the probability formulas. Its quantile is exactly $AC_\beta^{-1}(y)$.

Write $\|u\|_{\mathbb{R}/\mathbb{Z}} = \min_{m \in \mathbb{Z}} |u - m|$ and $d_n(\beta) = \|n\alpha + \beta - 1/2\|_{\mathbb{R}/\mathbb{Z}}$. For $0 \leq d \leq 1/2$,

$$4d \leq 2 \sin(\pi d) \leq 2\pi d. \quad (6)$$

A phase is called *exact* if $d_m(\beta) = 0$ for some $m \geq 1$, and *nonexact* otherwise. Irrationality permits at most one exact index. This terminology deliberately avoids imposing an unstated quantitative meaning on the word “nonresonant.”

Lemma 2.2 (Support and quantiles). *For irrational α , S_β has no atoms and its support is the whole interval $[0, \sum b_n]$. Its distribution function is continuous and strictly increasing on that interval. Thus $C_\beta^{-1}(y)$ is unambiguous for $0 < y < 1$.*

Proof. At least one coefficient is positive, so convolution with its uniform law eliminates atoms. Finite sums have full interval supports; their deterministic tail bound tends to zero. More explicitly, any interior point can be approximated by a finite sum with values in the interiors of its positive uniform intervals, while the remaining tail is made smaller than a prescribed neighborhood. This gives positive probability to every open subinterval of the claimed support. The stated strict increase follows. $\square$

3 The soft cutoff and the scalar baseline

Set

$$F(u) = \log \frac{u}{1 - e^{-u}} \quad (u > 0), \quad F(0) = 0, \quad h(z) = F(e^z), \quad h(-\infty) = 0. \quad (7)$$

The infinite product becomes

$$H_\beta(t) = \sum_{n \geq 1} h(t - \lambda n + f_n), \quad f_n = \log a_n \in [-\infty, \log 2]. \quad (8)$$

Lemma 3.1 (Elementary kernel estimates). *There is an absolute constant C such that*

$$0 \leq F(u) \leq u/2, \quad 0 \leq F'(u) \leq 1/2, \quad (9)$$

$$0 \leq h(z) - z_+ \leq Ce^{-|z|} \quad (z \in \mathbb{R}), \quad (10)$$

$$h'(z) = J(e^z), \quad J(u) = 1 - \frac{u}{e^u - 1}, \quad (11)$$

$$J(u) - uJ'(u) = V(u), \quad V(u) = 1 - \frac{u^2 e^u}{(e^u - 1)^2}. \quad (12)$$

At $u = 0$, $J(0) = V(0) = 0$. For $u \geq 0$, $0 \leq J(u) \leq 1$, $0 \leq V(u) \leq 1$, and

$$J(u) = O(u), \quad V(u) = O(u^2) \quad (u \downarrow 0), \quad J(u), V(u) \longrightarrow 1 \quad (u \rightarrow \infty)$$

with exponentially small errors at infinity, up to polynomial factors.

Proof. $F(u)$ is the negative logarithm of the Laplace transform of $\text{Unif}[0, 1]$. Jensen's inequality gives $F(u) \leq u/2$. Its derivative is the mean of this uniform variable under exponential tilt; it lies between 0 and its untilted mean $1/2$, since the derivative of that mean is the negative tilted variance. For $z \leq 0$, $h(z) \leq e^z/2$. For $z \geq 0$,

$$h(z) - z = -\log(1 - e^{-e^z}) \geq 0;$$

this is bounded by Ce^{-z} , with a larger constant covering $z = 0$. Differentiation gives (11) and (12).

Under tilt, uV_1 has density $e^{-u}/(1 - e^{-u})$ on $[0, u]$. Its mean and variance are $J(u)$ and $V(u)$, respectively. This proves nonnegativity. The displayed formulas prove the upper bounds by 1; alternatively the mean is at most the mean of an unconditioned unit exponential. Taylor expansion at zero and the explicit formulas at infinity prove the remaining estimates. $\square$

Define, for $t \geq 0$,

$$B_\lambda(t) = \sum_{n \geq 1} (t - \lambda n)_+. \quad (13)$$

Writing $N = \lfloor t/\lambda \rfloor$ and $\eta = \{t/\lambda\}$ gives the *exact* identity

$$B_\lambda(t) = Nt - \frac{\lambda N(N+1)}{2} = \frac{t^2}{2\lambda} - \frac{t}{2} + \frac{\lambda}{2}\eta(1-\eta). \quad (14)$$

Thus the linear term $-t/2$ is forced by the indexing $n \geq 1$; it is not a resonance effect.

For later use, the kernel bounds imply

$$\sum_{n \geq 1} h(t - an) = B_a(t) + O_\lambda(1) \quad \text{uniformly for } a \geq \lambda, \quad (15)$$

since the exponentially decaying errors in (10) can be summed over any lattice with spacing at least λ . They also give the useful truncation certificate

$$0 \leq H_\beta(t) - \sum_{n=1}^M F(e^t b_n) \leq \frac{e^t q^{M+1}}{1-q}. \quad (16)$$

The same bound holds for the omitted part of $H'_\beta(t)$, because $J(u) = uF'(u) \leq u/2$. A safe variance-tail bound is $4e^{2t}q^{2(M+1)}/(1-q^2)$, using $V(u) \leq u^2$.

4 The universal leading law for every irrational rotation

Theorem 4.1 (Phase-uniform irrational universality). *For every fixed irrational α and every $\lambda > 0$, uniformly in $\beta \in \mathbb{R}/\mathbb{Z}$,*

$$H_\beta(t) = \frac{t^2}{2\lambda} + o(t^2), \quad t \rightarrow \infty, \quad (17)$$

$$\log C_\beta(x) = -\frac{\log^2(1/x)}{2\lambda} + o(\log^2(1/x)), \quad x \downarrow 0, \quad (18)$$

$$-\log C_\beta^{-1}(y) \sim \sqrt{2\lambda \log(1/y)}, \quad y \downarrow 0. \quad (19)$$

The little-o terms need not be uniform over irrational α .

Proof. Since $a_n \leq 2$, monotonicity of h and (15) imply

$$H_\beta(t) \leq B_\lambda(t + \log 2) + O_\lambda(1) = \frac{t^2}{2\lambda} + O_\lambda(t).$$

For the lower bound, fix $0 < \varepsilon < 1$. The set $E_\varepsilon = \{u \in \mathbb{R}/\mathbb{Z} : 2|\cos \pi u| < \varepsilon\}$ is an interval about $1/2$, of length tending to zero as $\varepsilon \downarrow 0$. Irrational rotation gives

$$\#\{1 \leq n \leq N : n\alpha + \beta \in E_\varepsilon\} = |E_\varepsilon|N + o(N) \quad (20)$$

uniformly in β . To see the uniformity directly, a nonconstant Fourier mode has average bounded by $2/(N|1 - e^{2\pi i k \alpha}|)$, independently of the starting phase. Approximate the interval indicator from above and below by continuous periodic functions and then by finite trigonometric polynomials.

Retain the terms with $a_n \geq \varepsilon$, and put $N = \lfloor (t + \log \varepsilon)/\lambda \rfloor$. The inequality $h(z) \geq z_+$ gives

$$H_\beta(t) \geq B_\lambda(t + \log \varepsilon) - (t + |\log \varepsilon|)(|E_\varepsilon|N + o(N)).$$

Divide by t^2 , first let $t \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. This proves (17) uniformly in phase.

Set $L = \log(1/x)$. Chernoff's inequality at $s = 1/x$ gives

$$C_\beta(x) \leq \exp(1 - H_\beta(L)). \quad (21)$$

For a phase-independent lower bound, couple $S_\beta \leq S^* = 2 \sum q^n V_n$. Choose

$$N = \left\lceil \frac{L + \log(4/(1-q))}{\lambda} \right\rceil.$$

The deterministic tail of S^* after N is at most $x/2$. For sufficiently large L , each threshold $x/(4Nq^n)$, $1 \leq n \leq N$, lies in $(0, 1)$. On the event $V_n \leq x/(4Nq^n)$ for all these indices, the first N terms contribute at most $x/2$. Independence therefore gives

$$\begin{aligned} \log C_\beta(x) &\geq -NL - N \log(4N) + \frac{\lambda N(N+1)}{2} \\ &= -\frac{L^2}{2\lambda} - \frac{L \log L}{\lambda} + O_\lambda(L). \end{aligned}$$

Together with (21), this proves (18). Monotone inversion of the two uniform quadratic bounds proves (19); Lemma 2.2 supplies the inverse. $\square$

Proposition 4.2 (Rational comparison). *Let $\alpha = p/r$ be reduced, and let ν of the r periodic amplitudes $a_1, \dots, a_r$ be positive. If $\nu > 0$, then*

$$H_\beta(t) \sim \frac{\nu}{r} \frac{t^2}{2\lambda}, \quad \log C_\beta(x) \sim -\frac{\nu}{r} \frac{\log^2(1/x)}{2\lambda}.$$

For $r \geq 2$, a generic phase has $\nu = r$, while a phase orthogonal to one orbit direction has $\nu = r - 1$.

Proof. For each positive residue j , its contribution is $\sum_{k \geq 0} h(t - \lambda j - \lambda r k + \log a_j)$, whose leading term is $t^2/(2\lambda r)$ by the scalar cutoff. Sum over the ν residues. Chernoff gives the probability upper bound. The finite-coordinate lower-bound event used above now needs only the positive indices: their count through N is $\nu N/r + O_r(1)$ and their index sum is $\nu N^2/(2r) + O_r(N)$. The fixed nonzero amplitude factors change the logarithm by only $O_{\alpha, \beta, \lambda}(N)$, proving the lower bound. Distinct residues yield distinct points modulo one, so at most one amplitude vanishes when $r \geq 2$. $\square$

This rational/irrational difference is relevant to the Liouville construction: a very accurate rational approximation can temporarily suppress a positive fraction of the effective digits, although that fraction tends to zero along increasing rational denominators.

5 A uniform trimmed rotation estimate

For the next five sections impose the bounded-type hypothesis

$$\|m\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{\kappa}{m} \quad (m \geq 1) \quad \text{for some fixed } \kappa > 0. \quad (\text{BT})$$

This is the usual badly approximable condition, equivalent to bounded continued-fraction partial quotients. Constants may depend on α (or a chosen bound for its partial quotients) and λ , but never on β unless explicitly indicated.

Lemma 5.1 (Uniform interval discrepancy). *Under (BT), there is D_α such that, for every interval $I \subset \mathbb{R}/\mathbb{Z}$, $N \geq 1$, and $\beta \in \mathbb{R}/\mathbb{Z}$,*

$$|\#\{1 \leq n \leq N : n\alpha + \beta \in I\} - N|I|| \leq D_\alpha \log(2N).$$

Consequently, for a periodic function g of bounded variation,

$$\left| \sum_{n=1}^N g(n\alpha + \beta) - N \int_{\mathbb{R}} g \right| \leq D_\alpha \text{Var}(g) \log(2N), \quad (22)$$

with a harmless change of the constant.

Proof. Let p_j/Q_j be a continued-fraction convergent. The first Q_j points of an orbit block differ, after a permutation, from a translated equally spaced Q_j -point grid by less than $1/Q_{j+1} \leq 1/Q_j$ on the circle. Indeed, $|\alpha - p_j/Q_j| < 1/(Q_j Q_{j+1})$, and the accumulated error over a block of Q_j successive indices is at most $1/Q_{j+1}$ after fixing its starting phase. The grid count in an interval differs from $Q_j|I|$ by at most two. Only a bounded number of grid points can lie within $1/Q_j$ of the interval's endpoints. Thus a block of length Q_j has an absolute counting error bounded by a universal constant, uniformly in its starting phase.

If all partial quotients are bounded by A , the greedy continued-fraction expansion of N decomposes its first N orbit points into at most $A + 1$ blocks of each convergent length. The denominators grow at least as fast as the Fibonacci denominators, so there are $O(\log(2N))$ different lengths. Adding the block errors proves the interval estimate. Finally, apply Stieltjes integration by parts to the signed counting measure minus N times Lebesgue measure; its total mass is zero and its interval cumulative function is bounded by the estimate just proved. This gives (22). $\square$

Lemma 5.2 (One exceptional term suffices). *Let $f(u) = \log(2|\cos \pi u|)$, with value $-\infty$ at $1/2$. For $N \geq 2$, choose $m \in \{1, \dots, N\}$ minimizing $d_n = \|n\alpha + \beta - 1/2\|_{\mathbb{R}/\mathbb{Z}}$. Under (BT),*

$$d_n \geq \frac{\kappa}{2N} \quad (n \neq m), \quad (23)$$

$$f(n\alpha + \beta) \geq -\log N - C_\alpha \quad (n \neq m), \quad (24)$$

$$\left| \sum_{\substack{1 \leq n \leq N \\ n \neq m}} f(n\alpha + \beta) \right| \leq C_\alpha \log^2(2N). \quad (25)$$

The sum in (25) is finite even at an exact phase, since its only possible infinite term was removed.

Proof. Any two of the first N orbit points are separated by at least κ/N . If $d_m \geq \kappa/(2N)$, the first assertion is immediate. Otherwise the triangle inequality and the separation bound give $d_n \geq \kappa/N - d_m > \kappa/(2N)$ for every $n \neq m$. Inequality (6) gives (24).

Choose a constant C_0 large enough that $\tau_N = -\log N - C_0$ is no larger than the lower bound in (24), and put $g_N = \max(f, \tau_N)$. Its variation on the circle is $O_\alpha(\log(2N))$. Also

$$\int_{\mathbb{R}} / \mathbb{Z} f = 0, \quad 0 \leq \int_{\mathbb{R}} / \mathbb{Z} (g_N - f) = O_\alpha(1/N). \quad (26)$$

For completeness, the mean-zero identity follows from $\int_0^{\pi/2} \log \sin u \, du = -(\pi/2) \log 2$ and reflection. The integral identity for sine is obtained by adding its sine and cosine versions and substituting $2u$ in their product. The truncation correction is supported where the distance to $1/2$ is $O(1/N)$; integration of $\log(c/(Nd))$ over such an interval is $O(1/N)$, proving the second claim.

Equation (22) now gives $\sum_{n \leq N} g_N(n\alpha + \beta) = O_\alpha(\log^2(2N))$. Except at m , g_N agrees with f ; and $|g_N(m\alpha + \beta)| = O_\alpha(\log(2N))$. Subtract this one finite term. This proves (25). $\square$

The uniformity in this lemma is important. An estimate for a fixed unshifted Sudler product would not, without an additional argument, control a phase chosen extremely close to one of the first N singular orbit locations.

6 The resonance profile and the sharp Laplace law

Definition 6.1 (Truncated resonance profile). Let

$$k_n(\beta) = (-\log a_n(\beta))_+ \in [0, \infty], \quad R_\beta(t) = \max_{n \geq 1} \min\{k_n(\beta), (t - \lambda n)_+\}. \quad (27)$$

The conventions are $\min\{\infty, u\} = u$ for finite u and $R_\beta(t) = 0$ for $0 \leq t < \lambda$. Only finitely many indices contribute when t is fixed.

Each function in the maximum grows with slope 1 after its starting time λn until it reaches height k_n , after which it stays constant. Thus

$$0 \leq R_\beta(t) \leq t, \quad 0 \leq R_\beta(t_2) - R_\beta(t_1) \leq t_2 - t_1 \quad (t_2 \geq t_1). \quad (28)$$

This elementary finite maximum, rather than an untruncated singular sum, is the correction that survives uniformly over arbitrary phases.

Theorem 6.2 (Uniform profile expansion). Under (BT), as $t \rightarrow \infty$,

$$H_\beta(t) = B_\lambda(t) - R_\beta(t) + O_{\alpha,\lambda}(\log^2(2+t)) \quad (29)$$

uniformly for every $\beta \in \mathbb{R}/\mathbb{Z}$, including exact phases. Equivalently,

$$\log \mathcal{L}_\beta(e^t) = -\frac{t^2}{2\lambda} + \frac{t}{2} + R_\beta(t) + O_{\alpha,\lambda}(\log^2(2+t)). \quad (30)$$

Proof. Let $N = \lfloor t/\lambda \rfloor$ and choose the exceptional index m from Lemma 5.2. Set $a = t - \lambda m \geq 0$. First consider all $n \leq N$ other than m . Their f_n lie between $-\log N - C_\alpha$ and $\log 2$. Except in a final interval of $O_{\alpha,\lambda}(\log(2N))$ indices, the quantity $t - \lambda n + f_n$ is at least $3 \log(2N)$. On that early interval, (10) permits replacing $h(t - \lambda n + f_n)$ by $t - \lambda n + f_n$ with a total error $O_{\alpha,\lambda}(1)$. On the final interval both the negative part of this linear expression and the soft-cutoff error are $O_{\alpha,\lambda}(\log(2N))$ per term. Consequently,

$$\begin{aligned} \sum_{\substack{n \leq N \\ n \neq m}} h(t - \lambda n + f_n) &= \sum_{\substack{n \leq N \\ n \neq m}} (t - \lambda n) + \sum_{\substack{n \leq N \\ n \neq m}} f_n + O_{\alpha,\lambda}(\log^2(2N)) \\ &= B_\lambda(t) - a + O_{\alpha,\lambda}(\log^2(2N)). \end{aligned} \quad (31)$$

For $n > N$, $h(t - \lambda n + f_n) \leq e^{t - \lambda n}$, so the total tail is $O_\lambda(1)$.

If $f_m \leq 0$, including $f_m = -\infty$, the cutoff estimate gives

$$h(a + f_m) - a = -\min\{k_m, a\} + O(1).$$

For $f_m > 0$, the same statement holds with $k_m = 0$ and an $O(1)$ error, since $f_m \leq \log 2$ and h is 1-Lipschitz. Combining this with (31) yields

$$H_\beta(t) = B_\lambda(t) - \min\{k_m, t - \lambda m\} + O_{\alpha,\lambda}(\log^2(2N)).$$

All other indices $n \leq N$ have $k_n \leq \log N + C_\alpha$. Thus replacing the exceptional term by the maximum $R_\beta(t)$ changes the expression by at most $O_\alpha(\log(2N))$. Equation (14) finishes the proof. $\square$

Corollary 6.3 (Resolution of the displayed source estimate). *Under (BT), uniformly in phase,*

$$\log \mathcal{L}_\beta(s) = -\frac{(\log s)^2}{2\lambda} + O_{\alpha,\lambda}(\log s).$$

In particular, the estimate (1) holds, with no phase exclusion at all.

Proof. Use $0 \leq R_\beta(t) \leq t$ and $\log^2 t = o(t)$ in Theorem 6.2. $\square$

Corollary 6.4 (A quantitative nonresonance condition). *Suppose that, in addition to (BT), there are $c > 0$ and $\tau > 0$ with*

$$\|n\alpha + \beta - 1/2\|_{\mathbb{R}/\mathbb{Z}} \geq cn^{-\tau} \quad (n \geq 1). \quad (32)$$

Then $R_\beta(t) = O_{c,\tau,\lambda}(\log(2+t))$ and

$$\log \mathcal{L}_\beta(e^t) = -\frac{t^2}{2\lambda} + \frac{t}{2} + O_{\alpha,\lambda,c,\tau}(\log^2(2+t)).$$

Proof. Inequality (6) gives $k_n \leq \tau \log n + C_c$. In (27), only $n \leq t/\lambda$ matter. $\square$

Remark 6.5 (Why the untruncated cocycle is not the uniform correction). The source's cocycle is $\sum_{n \leq N} f_n$. A single f_m can be arbitrarily negative, or $-\infty$, while its actual Laplace contribution stays between 0 and a quantity of order $t - \lambda m$. The minimum in (27) is precisely this saturation. Before saturation, a small digit is effectively missing; after saturation, its full logarithmic amplitude enters. The formula is therefore compatible with continuity of the Laplace transform at a phase where a digit becomes exactly zero.

7 A relative saddle formula from tilted gamma blocks

A logarithmic Laplace estimate by itself is not a relative asymptotic for a small probability. This section supplies the missing local argument. For $s = e^t$, define the tilted probability measure

$$\frac{d\mathbb{P}_t}{d\mathbb{P}} = \exp(-sS_\beta + H_\beta(t)). \quad (33)$$

Expectations and variances under this measure are denoted by $\mathbb{E}_t$ and Var_t . Set

$$m_\beta(t) = H'_\beta(t), \quad \mathcal{V}_\beta(t) = H'_\beta(t) - H''_\beta(t) = s^2 \text{Var}_t(S_\beta). \quad (34)$$

The variance is strictly positive.

Lemma 7.1 (Tilted summands and their effective number). *Under (BT), uniformly in β as $t \rightarrow \infty$,*

$$H'_\beta(t) = \frac{t}{\lambda} + O_{\alpha,\lambda}(\log(2+t)), \quad \mathcal{V}_\beta(t) = \frac{t}{\lambda} + O_{\alpha,\lambda}(\log(2+t)). \quad (35)$$

More specifically, write $u_n = sb_n$ and $W_n = sb_n V_n$ under $\mathbb{P}_t$. The nonzero W_n are independent unit exponential variables conditioned to lie in $[0, u_n]$. Their mean and variance are $J(u_n)$ and $V(u_n)$. If $N = \lfloor t/\lambda \rfloor$, then all but one of the indices

$$1 \leq n \leq N - C_{\alpha,\lambda} \log(2N)$$

have $u_n \geq N^3$, after increasing the constant and taking N large. The sum of the means and the sum of the variances of all remaining terms are $O_{\alpha,\lambda}(\log(2N))$.

Proof. Tilting a product of independent laws preserves independence: for any finite set of indices, integrate the complementary factors in (33) and cancel their Laplace transforms. A uniform variable on $[0, u]$ becomes a variable with density $e^{-w}/(1 - e^{-u})$ on that interval. The formulas for J and V therefore apply, and summing them proves (34). Termwise differentiation is also justified directly by the summable small- u bounds on any compact t interval.

Use the same exceptional index as in Lemma 5.2. For every other $n \leq N$, $a_n \geq e^{-C_\alpha}/N$. If $\lambda(N - n) \geq 4 \log N + C_\alpha$, then $u_n = e^{t-\lambda n} a_n \geq N^3$. There are $N - O_{\alpha,\lambda}(\log(2N))$ such indices. For them, both mean and variance are 1 with exponentially small errors. The remaining indices through N number $O_{\alpha,\lambda}(\log(2N))$, and each has mean and variance at most 1. For $n > N$, the bounds $J(u) = O(u)$ and $V(u) = O(u^2)$ give bounded geometric tails. Summing proves all assertions. $\square$

Lemma 7.2 (A gamma block gives the needed local limit). *Suppose G_M has the gamma density*

$$g_M(w) = \frac{w^{M-1} e^{-w}}{\Gamma(M)} \mathbf{1}_{w>0}, \quad M \rightarrow \infty,$$

and B_M is independent of G_M with $\text{Var}(B_M) = O(\log M)$. Let $a_M = M + \mathbb{E}B_M + o(1)$. Then

$$\mathbb{E} [e^{G_M+B_M-a_M} \mathbf{1}_{G_M+B_M \leq a_M}] = \frac{1 + o(1)}{\sqrt{2\pi M}}. \quad (36)$$

The conclusion is uniform in any auxiliary parameters for which the stated variance bound and $o(1)$ are uniform. It remains true if the law of $G_M + B_M$ is changed by total variation $o(M^{-1/2})$, provided the same centering a_M is used.

Proof. Stirling's formula implies

$$g_M(M + y) = \frac{1 + o(1)}{\sqrt{2\pi M}} \quad \text{uniformly for } |y| \leq M^{1/4}, \quad \sup_w g_M(w) \leq \frac{C}{\sqrt{M}}. \quad (37)$$

For example, substitute $w = M + y$ in the logarithm of the density; the quadratic term is $-y^2/(2M) = O(M^{-1/2})$ and the remaining terms tend to zero uniformly on that window. The supremum occurs at $M - 1$ when $M > 1$ and follows from the same formula.

Let $\tilde{g}_M$ be the density of $G_M + B_M$. Conditioning on B_M ,

$$\tilde{g}_M(a_M - v) = \mathbb{E} g_M(a_M - v - B_M).$$

Chebyshev's inequality gives $\mathbb{P}(|B_M - \mathbb{E}B_M| > M^{1/4}/3) = O(\log M/\sqrt{M}) = o(1)$. For $0 \leq v \leq M^{1/4}/3$, the complementary event puts the gamma density inside the window in (37), for large M . The exceptional event contributes $o(M^{-1/2})$ by the global density bound. Thus $\tilde{g}_M(a_M - v) = (2\pi M)^{-1/2}(1 + o(1))$ uniformly on that v interval, and globally it is bounded by $C/\sqrt{M}$. Finally,

$$\mathbb{E}[e^{G_M+B_M-a_M} \mathbf{1}_{G_M+B_M \leq a_M}] = \int_0^\infty e^{-v} \tilde{g}_M(a_M - v) dv.$$

The tail after $M^{1/4}/3$ is negligible, proving the claim. The integrand defining the expectation is bounded by 1, so a total variation perturbation of the stated order changes it by $o(M^{-1/2})$. $\square$

Theorem 7.3 (Uniform relative cap saddle). *Assume (BT). For sufficiently small $x > 0$, let $t_x = t_x(\beta)$ be the unique solution of*

$$e^{t_x} x = H'_\beta(t_x). \quad (38)$$

Then, uniformly in $\beta \in \mathbb{R}/\mathbb{Z}$ as $x \downarrow 0$,

$$C_\beta(x) = \frac{\exp(e^{t_x} x - H_\beta(t_x))}{\sqrt{2\pi\mathcal{V}_\beta(t_x)}} (1 + o(1)). \quad (39)$$

Proof. The tilted mean $\mathbb{E}_s S_\beta$ is strictly decreasing in s , because its derivative is $-\text{Var}_s(S_\beta) < 0$. It tends to zero as $s \rightarrow \infty$, as also follows from Lemma 7.1. Its value at $s = 0$ is positive. For irrational α , two successive cosine factors cannot both vanish; compactness in β gives a positive uniform lower bound for the sum of their absolute values. The first two coefficients therefore give a positive uniform lower bound for $\mathbb{E} S_\beta$. It follows that for all sufficiently small x , the equation $\mathbb{E}_s S_\beta = x$ has a unique solution. That solution has $t_x \rightarrow \infty$ uniformly: for bounded s , positivity and continuity in β , again using the first two coefficients, bound the tilted mean away from zero. Equation (38) is its logarithmic form.

We now prove the required local statement at an arbitrary large t . Use the early indices of Lemma 7.1, excluding its possible exceptional index. Let their number be $M = N - O_{\alpha,\lambda}(\log(2N))$. Each early summand is a unit exponential conditioned to be at most u_n , where $u_n \geq N^3$. The total variation distance between that conditioned law and the unconditioned exponential law is e^{-u_n} . A product coupling therefore replaces the entire early sum by an independent gamma variable G_M with total variation error at most $N e^{-N^3}$.

Let B_M be the sum of all the other tilted summands. It is independent of the replaced early block and has mean and variance $O_{\alpha,\lambda}(\log(2N))$. The replacement changes the mean of the early block by at most

$$\sum_{\text{early } n} \frac{u_n}{e^{u_n} - 1} = O(N^4 e^{-N^3}) = o(1).$$

Consequently the original dimensionless sum $W = s S_\beta$ has center $m_\beta(t) = M + \mathbb{E} B_M + o(1)$ and its law is exponentially close in total variation to $G_M + B_M$. Lemma 7.2 yields

$$\mathbb{E}_t[e^{W - m_\beta(t)} \mathbf{1}_{W \leq m_\beta(t)}] = \frac{1 + o(1)}{\sqrt{2\pi\mathcal{V}_\beta(t)}}, \quad (40)$$

since $M \sim \mathcal{V}_\beta(t)$ uniformly. This proves more than a weak central limit theorem: it gives the exponentially weighted, one-sided local estimate needed for a relative probability.

At $t = t_x$, $s x = m_\beta(t_x)$. Reversing the tilt gives the exact identity

$$C_\beta(x) = e^{s x - H_\beta(t_x)} \mathbb{E}_{t_x}[e^{s S_\beta - s x} \mathbf{1}_{s S_\beta \leq s x}].$$

Insert (40). Every preceding estimate is phase-uniform, so the relative $o(1)$ in (39) is phase-uniform as well. $\square$

Remark 7.4 (Scope of the relative formula). Theorem 7.3 retains the *exact* functions H_β and $\mathcal{V}_\beta$ at the exact saddle. Replacing H_β by an approximation with $O(\log^2 t)$ error does not preserve a relative $1 + o(1)$ probability estimate. It does preserve the logarithmic expansion below, and that logarithmic accuracy is enough to give a relative inverse-quantile formula.

8 Three-term cap asymptotics and inverse quantiles

Define the constant

$$c_\lambda = \frac{1 + \log \lambda}{\lambda} + \frac{1}{2}. \quad (41)$$

For small $x > 0$, let

$$L = \log(1/x), \quad T = -W_{-1}(-\lambda x), \quad (42)$$

where W_{-1} is the real lower branch of Lambert's function. Equivalently, $T > 1$ is the large solution of $Te^{-T} = \lambda x$. Thus

$$T = L + \log L - \log \lambda + O_\lambda(\log L/L). \quad (43)$$

No complex branch convention is involved.

Theorem 8.1 (Phase-corrected cap probability). *Under (BT), uniformly in phase,*

$$t_x = T + O_{\alpha,\lambda}(\log T/T). \quad (44)$$

Moreover,

$$\log C_\beta(x) = -\frac{T^2}{2\lambda} + \left(\frac{1}{2} + \frac{1}{\lambda}\right)T + R_\beta(T) + O_{\alpha,\lambda}(\log^2 T), \quad (45)$$

$$= -\frac{L^2}{2\lambda} - \frac{L \log L}{\lambda} + c_\lambda L + R_\beta(L) + O_{\alpha,\lambda}(\log^2 L). \quad (46)$$

Proof. At the saddle, Lemma 7.1 gives $x = e^{-t_x}(t_x/\lambda)(1 + O_{\alpha,\lambda}(\log t_x/t_x))$. Taking logarithms yields

$$t_x - \log t_x = L - \log \lambda + O_{\alpha,\lambda}(\log t_x/t_x).$$

First this implies $t_x \sim L$ uniformly. Since $T - \log T = L - \log \lambda$ and the derivative of $u - \log u$ is bounded below by $1/2$ for $u \geq 2$, it gives (44). Equation (43) follows either by one substitution in the same implicit equation or by the identical derivative comparison.

Taking logarithms of the relative formula (39) and inserting Theorem 6.2 and Lemma 7.1 gives

$$\log C_\beta(x) = -\frac{t_x^2}{2\lambda} + \left(\frac{1}{2} + \frac{1}{\lambda}\right)t_x + R_\beta(t_x) + O_{\alpha,\lambda}(\log^2 t_x).$$

The variance prefactor contributes only $O(\log t_x)$ here. Changing t_x to T changes the quadratic expression by $O_{\alpha,\lambda}(\log T)$; (28) bounds the change in the profile. This proves (45). Expand T using (43), and note that $R_\beta(T) - R_\beta(L) = O_\lambda(\log L)$. The coefficient of L is exactly the constant in (41), giving (46). $\square$

Theorem 8.2 (Uniform multiplicative inverse-quantile law). *Assume (BT). Put*

$$z = \sqrt{2\lambda \log(1/y)}, \quad Q_\lambda(y) = \frac{z}{\lambda} \exp(-1 - \lambda/2 - z). \quad (47)$$

Then, uniformly in β as $y \downarrow 0$,

$$C_\beta^{-1}(y) = Q_\lambda(y) \exp\left(-\frac{\lambda R_\beta(z)}{z}\right) \left(1 + O_{\alpha,\lambda}\left(\frac{\log^2 z}{z}\right)\right). \quad (48)$$

Equivalently,

$$-\log C_\beta^{-1}(y) = z - \log z + 1 + \log \lambda + \frac{\lambda}{2} + \frac{\lambda R_\beta(z)}{z} + O_{\alpha,\lambda}(\log^2 z/z). \quad (49)$$

Proof. Let $M = \log(1/y)$, and consider

$$E_\beta(L) = \frac{L^2}{2\lambda} + \frac{L \log L}{\lambda} - c_\lambda L - R_\beta(L).$$

Equation (46) says $-\log C_\beta(e^{-L}) = E_\beta(L) + O_{\alpha,\lambda}(\log^2 L)$. Set

$$L_0 = z - \log z + \lambda c_\lambda + \frac{\lambda R_\beta(z)}{z}.$$

Since $R_\beta(z) \leq z$, $L_0 = z + O_\lambda(\log z)$ uniformly. Expanding the elementary terms, and using the Lipschitz bound for R_β , gives $E_\beta(L_0) = M + O_{\alpha,\lambda}(\log^2 z)$. For L near L_0 , the elementary part of E_β has slope at least $z/(2\lambda)$ for sufficiently large z , while the profile has Lipschitz constant 1. Hence increasing L by $K \log^2 z/z$ increases $E_\beta(L)$ by more than the possible error when K is sufficiently large; decreasing L has the reverse effect. Monotonicity of C_β brackets its inverse between these two choices of L . Thus $-\log C_\beta^{-1}(y) = L_0 + O_{\alpha,\lambda}(\log^2 z/z)$. Insert $\lambda c_\lambda = 1 + \log \lambda + \lambda/2$ and exponentiate. Since $\log^2 z/z \rightarrow 0$, exponentiating the error gives the relative error in (48). $\square$

Corollary 8.3 (A common asymptotic and a universal multiplicative range). *Under (BT), condition (32) implies*

$$C_\beta^{-1}(y) = Q_\lambda(y) \left(1 + O_{\alpha,\lambda,c,\tau}(\log^2 z/z)\right).$$

For arbitrary phases, the ratio $C_\beta^{-1}(y)/Q_\lambda(y)$ is asymptotically confined to $[q, 1]$, in the sense that its distance from this interval tends to zero uniformly in β .

Proof. Apply Corollary 6.4 in (48) for the first statement, and use $0 \leq R_\beta(z)/z \leq 1$ for the second. $\square$

9 Phase classification: measure, category, and quantile oscillation

For a nonexact phase define its inhomogeneous exponential approximation exponent by

$$\mu(\alpha, \beta) = \limsup_{n \rightarrow \infty} \frac{k_n(\beta)}{n} \in [0, \infty]. \quad (50)$$

By (6), replacing k_n by $(-\log d_n)_+$ leaves the exponent unchanged.

Lemma 9.1 (Optimization of the profile). *For any irrational α and any nonexact phase,*

$$\limsup_{t \rightarrow \infty} \frac{R_\beta(t)}{t} = \frac{\mu(\alpha, \beta)}{\lambda + \mu(\alpha, \beta)}, \quad (51)$$

where the right side is 1 when $\mu = \infty$. Under (BT), one also has

$$\liminf_{t \rightarrow \infty} \frac{R_\beta(t)}{t} = 0. \quad (52)$$

For an exact phase, instead, $R_\beta(t)/t \rightarrow 1$.

Proof. For a fixed n with finite k_n , the largest value of $\min\{k_n, (t - \lambda n)_+\}/t$ is $k_n/(\lambda n + k_n)$, attained at $t = \lambda n + k_n$. For finite μ , eventually $k_n \leq (\mu + \varepsilon)n$, giving the upper bound $(\mu + \varepsilon)/(\lambda + \mu + \varepsilon)$ for all sufficiently late indices. The finitely many earlier indices have bounded heights and contribute $o(1)$ after division by t . Indices with k_n/n approaching its limsup give the matching lower bound at their optimizing times. The case $\mu = \infty$ follows the same way from a sequence with $k_n/n \rightarrow \infty$; the case $\mu = 0$ follows already from the upper bound.

For (52), note that the sequence k_n is unbounded by density of the irrational orbit. At a nonexact phase every term is finite, so it has infinitely many strict record indices m_j . Ignore initial records with height zero. Writing $d_j = d_{m_j}$, the next record has $d_{j+1} < d_j$. The two orbit points therefore satisfy

$$\frac{\kappa}{m_{j+1} - m_j} \leq \|(m_{j+1} - m_j)\alpha\|_{\mathbb{R}/\mathbb{Z}} \leq d_j + d_{j+1} \leq 2d_j \leq \frac{1}{2}e^{-k_{m_j}},$$

where the last inequality follows from $4d_j \leq a_{m_j} = e^{-k_{m_j}}$. Hence $k_{m_j} \leq \log m_{j+1} + C_\alpha$. At the time $t_j = \lambda(m_{j+1} - 1/2)$ the new record has not yet entered the profile, so $R_\beta(t_j) \leq k_{m_j}$ and $R_\beta(t_j)/t_j \rightarrow 0$. Finally, if $k_m = \infty$ for an exact index, then $R_\beta(t) \geq (t - \lambda m)_+$; combine this with $R_\beta(t) \leq t$. $\square$

Theorem 9.2 (Endpoint and inverse-quantile spectrum). *Assume (BT), let β be nonexact, and put $p_\beta = \mu(\alpha, \beta)/(\lambda + \mu(\alpha, \beta))$ with the preceding convention at infinity. Then*

$$\begin{aligned} \liminf_{t \rightarrow \infty} \frac{\log \mathcal{L}_\beta(e^t) + t^2/(2\lambda)}{t} &= \frac{1}{2}, \\ \limsup_{t \rightarrow \infty} \frac{\log \mathcal{L}_\beta(e^t) + t^2/(2\lambda)}{t} &= \frac{1}{2} + p_\beta, \end{aligned} \quad (53)$$

$$\liminf_{L \rightarrow \infty} \frac{\log C_\beta(e^{-L}) + L^2/(2\lambda) + (L \log L)/\lambda}{L} = c_\lambda, \quad (54)$$

$$\limsup_{L \rightarrow \infty} \frac{\log C_\beta(e^{-L}) + L^2/(2\lambda) + (L \log L)/\lambda}{L} = c_\lambda + p_\beta, \quad (55)$$

$$\begin{aligned} \limsup_{y \downarrow 0} \frac{C_\beta^{-1}(y)}{Q_\lambda(y)} &= 1, \\ \liminf_{y \downarrow 0} \frac{C_\beta^{-1}(y)}{Q_\lambda(y)} &= \exp(-\lambda p_\beta). \end{aligned} \quad (56)$$

In particular, $\mu = 0$ gives $C_\beta^{-1}(y) \sim Q_\lambda(y)$, while $\mu = \infty$ gives lower and upper quantile-ratio limits q and 1. For an exact phase, the quantile ratio converges to q .

Proof. Divide (30) and (46) by their large variables; $\log^2 t/t \rightarrow 0$. Insert Lemma 9.1. Equation (48) and continuity and monotonicity of $u \mapsto \exp(-\lambda u)$ give the quantile limits. For an exact phase use the final conclusion of that lemma instead. $\square$

Theorem 9.3 (Almost every phase versus a thin residual set). *Fix a badly approximable irrational α and $\lambda > 0$. For Lebesgue-almost every phase β , condition (32) holds for every fixed $\tau > 1$ with a suitable constant, and*

$$C_\beta^{-1}(y) \sim Q_\lambda(y). \quad (57)$$

There is also a dense G_δ set of nonexact phases with $\mu = \infty$. This set has Hausdorff dimension zero and for all its phases

$$\liminf_{y \downarrow 0} \frac{C_\beta^{-1}(y)}{Q_\lambda(y)} = q, \quad \limsup_{y \downarrow 0} \frac{C_\beta^{-1}(y)}{Q_\lambda(y)} = 1. \quad (58)$$

Indeed, the entire set of nonexact phases with $\mu > 0$ has Hausdorff dimension zero.

Proof. For $\tau > 1$, the phase event $d_n(\beta) < n^{-\tau}$ is an interval of length at most $2n^{-\tau}$. Its lengths are summable. Borel–Cantelli implies that almost every phase belongs to only finitely many of these events. Remove the countable exact phases; the finitely many early positive distances can be included by decreasing the constant c . Taking a countable collection of rational exponents $\tau > 1$ gives the claim for every fixed real $\tau > 1$ as well, by choosing a smaller rational exponent. Corollary 8.3 proves (57).

For integers $K, N \geq 1$, let

$$U_{K,N} = \bigcup_{n \geq N} \{\beta \in \mathbb{R}/\mathbb{Z} : a_n(\beta) < e^{-Kn}\}.$$

Each is open and dense: it contains a neighborhood of every center $1/2 - n\alpha$ for $n \geq N$, and those centers are dense. The intersection of all $U_{K,N}$ is a dense G_δ set on which $\mu = \infty$. Intersecting further with the complement of the countable exact phases preserves the dense G_δ property. Theorem 9.2 then proves (58).

Finally, if $\mu > 0$, then for some integer $K \geq 1$ the inequality $a_n < e^{-n/K}$ holds infinitely often. By (6), each such event is contained in an interval of length $O(e^{-n/K})$. For every $d > 0$, the sum of the d th powers of these lengths is finite. Covering the limsup set by all intervals after an arbitrary index N , and then letting $N \rightarrow \infty$, makes its d -dimensional Hausdorff measure zero. A countable union over K still has Hausdorff dimension zero. $\square$

Corollary 9.4 (A precise obstruction to a small carrier). *For every fixed badly approximable α , there are nonexact phases, forming a dense G_δ set, such that*

$$\limsup_{t \rightarrow \infty} \frac{\log \mathcal{L}_\beta(e^t) + t^2/(2\lambda) - t/2}{t} = 1.$$

Thus at these phases the remainder after the universal quadratic and linear terms is neither bounded nor $O(\log t)$.

Proof. Use (53) with $\mu = \infty$ and subtract $1/2$. $\square$

Remark 9.5 (Interpretation of the source conjecture). The preceding corollary does not contradict the source's displayed estimate; that estimate holds in the stronger form of Corollary 6.3. It corrects the accompanying carrier expectation *if* the word nonresonant is taken to mean only that no factor is exactly zero. Under a quantitative interpretation such as (32), the exceptional profile is only logarithmic, and further control of the remaining $O(\log^2 t)$ term is an open refinement rather than a contradiction.

10 Liouville rotations with superlinear corrections

We now drop (BT) and fix the phase $\beta = 1/2$. Thus $a_n = 2|\sin \pi n\alpha|$. To isolate a genuine resonance effect rather than the automatic linear term, define the baseline-corrected defect

$$\Delta_\alpha(t) = B_\lambda(t) - H_{\alpha,1/2}(t). \quad (59)$$

The extra α subscript makes variation in the rotation explicit.

Lemma 10.1 (Uniform rational shadow and continuity bound). *For a reduced rational $p/r, r \geq 2$,*

$$\Delta_{p/r}(t) = B_{\lambda r}(t) + O_\lambda((t+1)\log(2r)) = \frac{t^2}{2\lambda r} + O_\lambda((t+1)\log(2r) + r). \quad (60)$$

The constants are independent of p and r . For any real α, α' ,

$$|H_{\alpha,1/2}(t) - H_{\alpha',1/2}(t)| \leq K_\lambda e^t |\alpha - \alpha'|, \quad K_\lambda = \frac{\pi q}{(1-q)^2}. \quad (61)$$

Proof. At $\alpha = p/r$, exactly the multiples of r vanish. For all other indices, multiplication by p permutes the nonzero residues, so

$$2 \sin(\pi/r) \leq a_n \leq 2, \quad 4/r \leq a_n \leq 2.$$

For $n \leq \lfloor t/\lambda \rfloor$, replacing the nonzero factor a_n by 1 changes its h value by at most $|\log a_n| \leq C \log(2r)$ because $0 \leq h' \leq 1$. The sum of all terms past that index is $O_\lambda(1)$ both before and after the replacement. Consequently,

$$H_{p/r,1/2}(t) = \sum_{n \geq 1} h(t - \lambda n) - \sum_{j \geq 1} h(t - \lambda r j) + O_\lambda((t+1) \log(2r)).$$

Use (15), which is uniform for lattice spacings $\lambda r \geq \lambda$, followed by (14).

For the continuity bound,

$$|2q^n |\sin \pi n \alpha| - 2q^n |\sin \pi n \alpha'|| \leq 2\pi n q^n |\alpha - \alpha'|.$$

Since F is $1/2$ -Lipschitz by Lemma 3.1, sum the resulting bounds for $F(e^t b_n)$ and use $\sum_{n \geq 1} n q^n = q/(1-q)^2$. $\square$

Theorem 10.2 (Dense Liouville families with arbitrarily near-quadratic spikes). *Fix $\lambda > 0$. There is a dense G_δ set of irrational Liouville numbers α such that the following holds simultaneously for every integer $k \geq 2$: there is a sequence $t_j \rightarrow \infty$ with*

$$\Delta_\alpha(t_j) \sim \frac{1}{2\lambda} t_j^{2-1/k}. \quad (62)$$

The phase can be fixed at $\beta = 1/2$ for the whole construction. In particular, the correction is larger than every fixed multiple of t along the $k = 2$ subsequence, even after subtracting the exact scalar baseline.

Proof. For integers $k \geq 2$ and $N \geq 1$, define

$$\mathcal{U}_{k,N} = \bigcup_{\substack{r \geq N, p \in \mathbb{Z} \\ (p,r)=1}} \left(\frac{p}{r} - e^{-2r^k}, \frac{p}{r} + e^{-2r^k} \right).$$

These sets are open and dense in $\mathbb{R}$, since reduced rationals with arbitrarily large denominators are dense. Their countable intersection $\mathcal{A} = \bigcap_{k \geq 2} \bigcap_{N \geq 1} \mathcal{U}_{k,N}$ is dense G_δ . A rational a/b cannot belong: for sufficiently large r , its distance to a different reduced p/r is at least $1/(br)$, which exceeds e^{-2r^k} . Every member is therefore irrational and admits approximations stronger than r^{-M} for every M , hence is Liouville.

Fix $\alpha \in \mathcal{A}$ and an integer $k \geq 2$. Choose reduced p_j/r_j with $r_j \rightarrow \infty$ and $|\alpha - p_j/r_j| < e^{-2r_j^k}$, and set $t_j = r_j^k$. By (61),

$$|\Delta_\alpha(t_j) - \Delta_{p_j/r_j}(t_j)| \leq K_\lambda e^{-t_j}.$$

The rational estimate gives

$$\Delta_{p_j/r_j}(t_j) = \frac{r_j^{2k-1}}{2\lambda} + O_\lambda(r_j^k \log(2r_j) + r_j).$$

The error is $o(r_j^{2k-1})$ because $k \geq 2$. Since $r_j^{2k-1} = t_j^{2-1/k}$, the desired asymptotic follows. The definition of $\mathcal{A}$ ensures that this works for every k , with different subsequences allowed. $\square$

Remark 10.3 (Consistency with irrational universality). For each fixed k , the exponent $2 - 1/k$ is strictly below 2. Thus these large corrections are compatible with $H_\alpha(t) \sim t^2/(2\lambda)$ from Theorem 4.1. The effective rational fraction of missing digits is $1/r_j$, which tends to zero, although its integrated contribution $t_j^2/(2\lambda r_j)$ can be much larger than a linear term.

Remark 10.4 (What is not transferred to Liouville cap quantiles). The relative saddle proof above uses a bounded-type separation estimate to produce a gamma block with only logarithmically many exceptions. It has not been extended here to the Liouville family. The Liouville result concerns Laplace corrections; the leading cap and inverse laws still follow from Theorem 4.1, but a sharp Liouville probability or quantile spike formula is not asserted.

11 Reproducible calculations and validation limits

The accompanying program `verify_endpoints.py` evaluates the elementary infinite sums directly in logarithmic coordinates. It requires Python and `mpmath`, uses no network access or random sampling, and writes its results to CSV and JSON. The mathematical proofs do not depend on these calculations. The tables below are finite examples, not evidence sufficient to infer an infinite-subsequence theorem.

11.1 Bounded-type and deliberately resonant phases

Take $q = 1/2$, $\lambda = \log 2$, and $\alpha = (\sqrt{5} - 1)/2$. Alongside $\beta = 0$, consider the phase

$$\beta_* = \frac{1}{2} - 20\alpha + e^{-120} \pmod{1}. \quad (63)$$

Its twentieth amplitude is $2 \sin(\pi e^{-120})$, so $k_{20} = 120 - \log(2\pi) + O(e^{-240})$. Before the corresponding height is reached, its contribution to the profile is $t - 20 \log 2$; after that it saturates. This is a controlled demonstration of one long ramp, not a construction of infinitely many ramps at a fixed phase. The latter is supplied by the Baire argument, not by the example.

Phase	t	$R_\beta(t)$	$H - B + R$	$H'/(t/\lambda)$
$\beta = 0$	40	3.186248	5.003670	0.994760
$\beta = 0$	80	4.629816	5.743629	0.995093
$\beta = 0$	120	4.629816	5.770088	0.998650
$\beta = 0$	160	4.629816	6.306261	0.997050
engineered phase	40	26.137056	5.767516	0.969153
engineered phase	80	66.137056	5.824267	0.986599
engineered phase	120	106.137056	6.067652	0.991524
engineered phase	160	118.162123	7.265925	0.998964

The residual $H - B + R$ is modest at the displayed scales, while omitting R would miss a correction exceeding 100 in the deliberately resonant case. The table does not establish an optimal residual order or numerical constants for the asymptotic theorems. The second ordinary phase $\beta = 1/2$ and additional values are retained in the CSV file.

11.2 Rational shadows for the Liouville construction

For $\alpha = 1/r$, $\beta = 1/2$, and $t = r^2$, the theorem predicts $\Delta_{1/r}(t) \sim t^2/(2\lambda r)$ as $r \rightarrow \infty$. Multiples of r are set to zero exactly in the code rather than by rounded sine evaluation.

r	$t = r^2$	$\Delta_{1/r}(t)$	$t^2/(2\lambda r)$	ratio
7	49	202.642024	247.422200	0.819013
13	169	1450.989793	1584.800502	0.915566
23	529	8410.452487	8776.635281	0.958278
41	1681	48661.591192	49715.992457	0.978792

For any irrational α' satisfying $|\alpha' - 1/r| < e^{-2r^2}$, Lemma 10.1 transfers the corresponding value of H with error at most $K_\lambda e^{-r^2}$. This continuity bound is analytic. The table alone does not exhibit one irrational number having all these approximations; the nested dense-open construction in Theorem 10.2 supplies suitable single numbers.

11.3 Kernel, precision, and saddle checks

The program checks the finite-sum and fractional-part versions of (14); numerically differentiates the kernel to check $h' = J$ and $J - uJ' = V$; tests one profile increment against its monotonicity and Lipschitz bounds; and recomputes the deliberately resonant example at higher precision. With the default 110 decimal digits and an additional 35 digits for that comparison, the absolute difference in $H_{\beta_*}(120)$ was approximately 5.51×10^{-65} .

For $\beta = 0$ and $x = e^{-20}$, deterministic bisection gave

$$T = 23.52455780924499 \dots, \quad t_x = 23.51292579548681 \dots,$$

with a relative residual in the saddle equation below 6×10^{-31} . The saddle expression for $\log C_0(e^{-20})$ evaluates to approximately -357.5854846402 . This last number is explicitly a *saddle approximation*: the program does not independently evaluate the exact cap CDF, so it supplies no finite- x relative error certificate for that number.

The analytic omitted-tail bounds in the main tables are below 10^{-40} . Rounding error is separate: calculations use high precision and a precision comparison, not interval arithmetic. The JSON report records these limits, the actual equation residual, and the checks performed. No Lean, Coq, or other kernel verification is claimed.

11.4 Dependency and formalization roadmap

The proof dependencies are short enough to separate into four layers. The distributional layer consists of the uniform-series construction, Laplace product, tilted summands, and scalar kernel inequalities. The arithmetic layer consists of the bounded-type spacing and interval-discrepancy bounds, followed by the trimmed singular sum. The analytic layer passes from that sum to the profile expansion and the gamma-block local limit. The final layer is real-variable inversion and the measure/category classification. The Liouville argument branches off after the kernel layer and uses rational comparison, continuity, and Baire category instead.

A practical formalization should first establish the deterministic finite profile and the kernel bounds, which also support certified computation. Next it should formalize the continued-fraction block discrepancy and the truncation integral. Only then is the infinite-product and probability infrastructure needed. The relative local-limit step should be formalized as the standalone gamma-convolution lemma rather than hidden inside an unproved appeal to a saddle-point method. A separate classical-analysis proof of monotone inverse bracketing suffices for the quantile theorem. These are proposed proof interfaces, not untested Lean code presented as working implementation.

12 Further research questions and topics

The questions below distinguish refinements of this article from broader problems requiring an additional literature check. In particular, a question about importing a sharper shifted Sudler estimate is not a claim that the corresponding estimate is unknown in all existing work.

Research question 12.1 (Sharp uniform residual after extracting the profile). Can the $O_{\alpha,\lambda}(\log^2 t)$ residual in (29) be replaced by $O_{\alpha,\lambda}(\log t)$, or by a smaller explicit arithmetic term plus a bounded remainder? First determine which available shifted Sudler-product estimates apply with constants uniform in phase after the nearest singular term has been removed. The target is the cap expansion, not a rediscovery of an existing unshifted product estimate.

Research question 12.2 (Renormalized carriers at quadratic rotations). For a quadratic irrational α and an explicitly specified arithmetic class of phases, identify the behavior of $H_{\beta}(t) - B_{\lambda}(t) + R_{\beta}(t)$ along

continued-fraction scales. Is the natural carrier periodic in a renormalization coordinate, quasiperiodic, or better described by an iterated functional relation? Any proposed bounded carrier should state the phase class and the variable in which boundedness is claimed.

Research question 12.3 (Effective finite-exponent phase realization). For a fixed badly approximable α and prescribed $0 < \mu < \infty$, construct or import a construction of nonexact phases with $\mu(\alpha, \beta) = \mu$, while controlling the optimizing subsequences. Theorem 9.2 then gives the corresponding quantile-ratio interval explicitly. An effective construction with error bounds would add more information than the conditional classification proved here.

Research question 12.4 (Sharp Liouville cap and inverse spikes). Extend the rational-shadow argument beyond the Laplace transform to a relative tilted local limit on the subsequences $t_j = r_j^k$. Determine the resulting deviations of $C_{\alpha,1/2}(x)$ and its inverse from their universal leading laws. A promising route is to separate the nearly vanished rational-residue terms from a large remaining gamma block; the required estimates must be uniform in the increasing denominator.

Research question 12.5 (Several singular directions and matrix cap laws). Replace one cosine amplitude by a finite family of projected rotating vectors, or by a periodic amplitude with several zeros of specified orders. Determine whether a sum of finitely many truncated resonance profiles gives a phase-uniform expansion. For genuine matrix dilations, investigate how multiple rotation frequencies, unequal contraction rates, and polynomial Jordan factors change both the quadratic coefficient and the $L \log L$ correction.

Research question 12.6 (Quantitative local errors and joint parameter limits). Derive explicit constants and rates in the gamma-block approximation and in (39). Then study a joint limit $x \downarrow 0$ and $q \uparrow 1$. The present constants are allowed to depend on λ , so the theorems cannot simply be used uniformly as $\lambda \downarrow 0$. Locating the transition between the cap regime and the nearly Gaussian central regime would connect endpoint and bulk asymptotics.

Research question 12.7 (Statistics under phase averaging). For a uniformly distributed phase, estimate the distribution of $R_\beta(t)$, its moments, and the tails of the quantile ratio at a finite scale. Almost-everywhere asymptotics and Hausdorff dimension zero do not supply quantitative finite-scale probabilities. The truncated nature of the profile makes this a bounded observable even at exact phases.

Research question 12.8 (Certified computation and proof-assistant integration). Implement interval-certified evaluation of H_β , H'_β , and $\mathcal{V}_\beta$, combined with a rigorous saddle bracket and explicit local-limit bounds. Exploit the finite maximum in (27) to avoid having to decide whether a computable phase is exactly singular. Formalize the kernel, discrepancy, trimming, and inverse-bracketing interfaces in the repository before attempting a monolithic formalization of the full saddle theorem.

13 Conclusions

The leading quadratic endpoint scale is much more robust than the singular cocycle alone suggests: every irrational rotation has it, uniformly in phase. Bounded type gives a stronger structural conclusion. After a deterministic scalar baseline is removed, a single truncated deepest-return profile controls the Laplace transform up to a squared-logarithmic error. This control is strong enough, through a relative tilted local limit, to determine a multiplicative inverse-quantile asymptotic.

The arithmetic distinction is not merely between rational, Diophantine, and Liouville angles. Even at one fixed badly approximable angle, the phase has its own exponential approximation exponent. It governs a real, explicitly quantified oscillation in extreme quantiles. Almost-everywhere universality coexists with a dense, dimension-zero exceptional set. Liouville angles introduce a further mechanism: long rational shadows suppress whole residue classes and produce near-quadratic corrections along selected subsequences.

Thus the two concrete repository questions lead to both positive theorems and a necessary qualification. The displayed Diophantine bound is valid in a stronger form, and Liouville spikes exist in a stronger baseline-corrected form. A bounded or logarithmic carrier, however, needs a quantitative phase hypothesis and cannot follow from absence of exact zeros alone.

A Source provenance, normalization checks, and package contents

The inspected repository is <https://github.com/VladimirReshetnikov/ProveIt>. The relevant report is located at

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/Matrix_Dilated_Fabius_Rvachev_Frontier_Report/matrix_dilated_fabius_rvachev.tex.
```

The pinned repository revision is `fbba58593dc0622aa914972896150d4848f935b5`, and the report blob is `143f6da778dfc0d5ab04133005ba6715a3cf155b`. The defining cap product appears in the section on directional caps; the two question labels are `conj:diophantine-endpoint` and `conj:liouville`, inspected in source lines approximately 1565–1610 of this revision. The report is a repository research document, not an independently refereed source of the new theorems in this article.

The present normalization uses $n \geq 1$, the factor $2|\cos \pi(n\alpha + \beta)|$, and the convention $q = e^{-\lambda}$. The logarithmic phase average is zero: $\int_{\mathbb{R}} / \mathbb{Z} \log(2|\cos \pi u|) \, du = 0$. Consequently the common quantile Q_λ corresponds to the scalar baseline with weights q^n , not to a silently rescaled scalar law with weights $2q^n$. A deterministic external scale multiplies every quantile by that scale, as in Proposition 2.1. Exact zero factors are omitted from products, assigned $k_n = \infty$ in profiles, and are handled separately from nonexact phases in the exponent classification.

The archive contains the self-contained LaTeX driver `article.tex`, its compiled `article.pdf`, the deterministic verification program, `requirements.txt`, a README with reproduction commands, the two CSV result files, their LaTeX table counterparts, a JSON verification report, and a machine-readable provenance record. No external repository source, font file, or unverified proof-assistant script is needed to build the article. The only LaTeX inclusions are the two small table files supplied in `data/`.

References

- [1] ProveIt repository research report. *Matrix-Dilated Fabius–Rvachev Laws: Infinite Box Splines, Thue–Morse Directional Derivatives, Bell–Bernoulli Tensor Resolvents, and Rotating-Zonoid q -Series*. Report dated 30 August 2026; inspected at the repository revision recorded in Appendix A. In particular, the directional-cap section and source labels `conj:diophantine-endpoint` and `conj:liouville`. https://github.com/VladimirReshetnikov/ProveIt/blob/fbba58593dc0622aa914972896150d4848f935b5/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/Matrix_Dilated_Fabius_Rvachev_Frontier_Report/matrix_dilated_fabius_rvachev.tex.
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